



AWS CLOUD COMPLETE TEXTBOOK

From Basics to Real-World Architecture

24 Chapters • 100+ Services • Diagrams & Use Cases

tech__.talks

1 Chapter 1

AWS BASICS

What is Cloud Computing?

Cloud computing is the on-demand delivery of IT resources — compute, storage, databases, networking — over the Internet with pay-as-you-go pricing. Instead of buying and maintaining physical servers, you rent capacity from a cloud provider.

Traditional IT vs Cloud Computing

Traditional: Buy servers → Wait weeks → Pay upfront → Maintain hardware
Cloud: Request online → Get in minutes → Pay per use → No hardware
Analogy: Like electricity — you don't build a power plant; you plug in and pay the bill.

Key Characteristics of Cloud Computing

- On-demand self-service — provision resources instantly, no human interaction
- Broad network access — accessible from any device over the Internet
- Resource pooling — shared infrastructure across multiple customers
- Rapid elasticity — scale up or down automatically
- Measured service — pay only for what you use

Why AWS?

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform. Launched in 2006, it offers over 200 fully featured services from data centers globally.

AWS Advantage	Detail
Market Leader	Over 30% global cloud market share
200+ Services	Compute, storage, AI, IoT, security, and more
Global Reach	30+ regions, 96+ Availability Zones worldwide
Pay-as-you-go	No upfront cost, no long-term commitment required
Security	Highest certifications — SOC 2, ISO 27001, HIPAA, FedRAMP
Community	Largest community, documentation, and partner ecosystem

Cloud Service Models

Cloud services are grouped into three models based on what the provider manages vs what you manage.

Responsibility Stack (bottom = provider manages, top = you manage)

SaaS ► Application + Data + Runtime + OS + Infra [Provider manages all]

PaaS ► Application + Data [You] Runtime + OS + Infra [Provider]
IaaS ► Application + Data + Runtime + OS [You] Infra [Provider]
Analogy: Pizza — Make at home (IaaS) / Take & Bake (PaaS) / Delivery (SaaS)

IaaS — Infrastructure as a Service

You manage: OS, runtime, middleware, applications, data. Provider manages: physical servers, networking, storage hardware.

- Examples: EC2, VPC, EBS, S3
- Use when: You need full control over the OS and environment

PaaS — Platform as a Service

You manage: applications and data only. Provider manages everything below — runtime, OS, infrastructure.

- Examples: Elastic Beanstalk, RDS, Lambda (partially)
- Use when: You want to focus on code, not infrastructure

SaaS — Software as a Service

You use the software; the provider manages everything. Access via browser or API.

- Examples: Gmail, Salesforce, AWS WorkMail
- Use when: You need a ready-to-use application

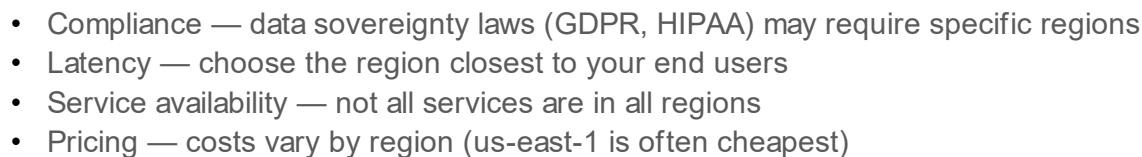
Cloud Deployment Models

Model	Description	AWS Example
Public Cloud	Resources owned by provider, shared via internet	Full AWS deployment
Private Cloud	Dedicated resources for one organization	AWS Outposts on-premises
Hybrid Cloud	Mix of on-premises + public cloud, connected securely	VPN / Direct Connect to AWS

 **Exam Tip:** Most AWS services are Public Cloud. Outposts brings AWS hardware to your data center (Private/Hybrid). Direct Connect provides dedicated private connectivity.

tech__talks

An AWS Region is a geographic area containing multiple isolated data centers. Each region is completely independent — data in one region does not automatically replicate to another.



An Availability Zone is one or more discrete data centers with redundant power, networking, and connectivity within a Region. AZs are physically separated (miles apart) but connected via low-latency private fiber.

Edge Locations are AWS data centers used by CloudFront (CDN) and Route 53 to cache content and serve DNS closer to end users. There are 400+ edge locations globally — many more than Regions.

Edge Location Role

User in Mumbai → Request hits nearby Edge Location → Cached content served

If not cached → Edge Location fetches from Origin (S3 / ALB) → Caches it

Result: Low latency for global users without running servers everywhere

Regional vs Global Services

Scope	Examples
Global Services	IAM, Route 53, CloudFront, WAF, Organizations
Regional Services	EC2, S3, RDS, VPC, Lambda, EKS, SQS, SNS

💡 *S3 buckets are created in a region, but bucket names are globally unique. IAM is global — one IAM policy applies across all regions.*

tech__talks

3 Chapter 3**AWS ACCOUNT & MANAGEMENT****AWS Account Structure**

Every AWS customer has one or more AWS accounts. An account is the fundamental boundary for billing, security, and resource isolation.

Root User

The root user is created when you open an AWS account. It has unrestricted access to all resources and billing.

- Email + password login (not access keys for daily use)
- Best practice: Enable MFA immediately, then NEVER use root for daily tasks
- Root-only tasks: Change account name, close account, change support plan

IAM User

Individual identities created within an account for people or applications. Each has its own credentials and permissions.

IAM Group

A collection of IAM users. Assign policies to groups, not individual users — easier to manage at scale.

AWS Organizations

AWS Organizations allows you to manage multiple AWS accounts centrally. You can group accounts into Organizational Units (OUs) and apply Service Control Policies (SCPs) to restrict what actions accounts can perform.

AWS Organizations Hierarchy	
Root (Management Account)	
└─ OU: Production	
└─ Account: prod-app	
└─ Account: prod-db	
└─ OU: Development	
└─ Account: dev-team	
└─ OU: Security	
└─ Account: security-audit	

Billing & Cost Management

Tool	Purpose
AWS Cost Explorer	Visualize and analyze spending trends

AWS Budgets	Set custom cost/usage alerts
Cost Allocation Tags	Tag resources to track costs by project/team
Consolidated Billing	Single bill for all accounts in an Organization
Savings Plans	Commit to usage for 1-3 years for discounts

tech__talks

Chapter 4

IDENTITY & ACCESS MANAGEMENT (IAM)

IAM Overview

IAM (Identity and Access Management) is AWS's free, global service for managing authentication (who are you?) and authorization (what can you do?). Every API call in AWS goes through IAM.

IAM Components	
Users	→ Long-term credentials for humans or apps
Groups	→ Collection of users sharing the same permissions
Roles	→ Temporary credentials assumed by services, apps, or federated users
Policies	→ JSON documents defining Allow/Deny actions on resources

IAM Policies

Policies are JSON documents attached to users, groups, or roles. They define what actions are allowed or denied on which AWS resources.

Example IAM Policy (Allow S3 Read)	
{	
"Version": "2012-10-17",	
"Statement": [{	
"Effect": "Allow",	
"Action": ["s3:GetObject", "s3:ListBucket"],	
"Resource": "arn:aws:s3:::my-bucket/*"	
}]	
}	

Managed Policies vs Inline Policies

Type	Description
AWS Managed	Pre-built by AWS, updated automatically (e.g., AdministratorAccess)
Customer Managed	Created by you, reusable across multiple identities
Inline Policy	Embedded directly into a single user/group/role — not reusable

IAM Roles

Roles provide temporary security credentials (via STS). Used by EC2 instances, Lambda functions, cross-account access, and federated identity.

- EC2 Role: EC2 instance assumes role → can access S3 without access keys in code
- Lambda Role: Lambda function assumes role → can write to DynamoDB
- Cross-account: Account A assumes role in Account B for access

MFA — Multi-Factor Authentication

MFA adds a second layer of security (something you know + something you have). Always enable MFA for root and admin users.

- Virtual MFA: Google Authenticator, Authy, AWS MFA app
- Hardware MFA: Physical TOTP device or FIDO security key

STS — Security Token Service

STS issues temporary credentials (access key ID + secret + session token) when a role is assumed. Credentials expire automatically — much safer than long-term access keys.

 *Best Practice: Never use root user for daily tasks. Never store access keys in code. Use IAM Roles for all AWS services. Rotate access keys every 90 days.*

tech__talks

5 Chapter 5

COMPUTE SERVICES

EC2 — Elastic Compute Cloud

EC2 provides resizable virtual servers (instances) in the cloud. You choose the OS, CPU, memory, storage, and network. EC2 is the backbone of AWS compute.

EC2 Instance Types

Family	Use Case
General Purpose (t3, m6i)	Web servers, dev environments, small databases
Compute Optimized (c6i, c7g)	Batch processing, media encoding, HPC
Memory Optimized (r6i, x2)	In-memory databases, real-time analytics
Storage Optimized (i3, d3)	High I/O databases, data warehousing
Accelerated (p4, g5, inf2)	ML training, GPU rendering, deep learning

AMI — Amazon Machine Image

An AMI is a template that contains the OS, application server, and applications needed to launch an EC2 instance. Think of it as a snapshot/blueprint for your server.

- AWS-provided: Amazon Linux 2023, Ubuntu, Windows Server
- Marketplace: Pre-configured AMIs from vendors (e.g., Bitnami WordPress)
- Custom: Create your own AMI from a configured instance

Auto Scaling

Auto Scaling automatically adjusts the number of EC2 instances in response to demand. It maintains application availability and reduces costs by scaling in during low traffic.

Auto Scaling Components	
Launch Template	→ Defines what to launch (AMI, instance type, security group)
Auto Scaling Group	→ Defines min/desired/max instances and subnet placement
Scaling Policy	→ When to scale (CPU > 70% → add instance, CPU < 30% → remove)
Health Check	→ Unhealthy instance → terminate → replace automatically

Elastic Load Balancer (ELB)

ELB automatically distributes incoming application traffic across multiple targets (EC2, containers, IPs) in multiple AZs for fault tolerance.

Type	Layer	Best For
------	-------	----------

ALB (Application)	Layer 7 (HTTP/HTTPS)	Web apps, microservices, path/host routing
NLB (Network)	Layer 4 (TCP/UDP)	Ultra-low latency, gaming, IoT, static IP
CLB (Classic)	Layer 4 & 7	Legacy — do not use for new workloads

Lambda — Serverless Compute

AWS Lambda runs your code in response to events without provisioning or managing servers. You only pay for the compute time consumed — billed per millisecond.

- Supports: Python, Node.js, Java, Go, Ruby, .NET, custom runtimes
- Max execution: 15 minutes per invocation
- Memory: 128 MB to 10 GB
- Triggers: API Gateway, S3, DynamoDB, SQS, EventBridge, CloudWatch

ECS & EKS — Container Services

Service	Description
ECS (Elastic Container Service)	AWS-native container orchestration. Run Docker containers on EC2 or Fargate
EKS (Elastic Kubernetes Service)	Managed Kubernetes. Run Kubernetes workloads without managing the control plane
Fargate	Serverless compute for containers — no EC2 management. Works with ECS and EKS

tech__talks

6 Chapter 6

STORAGE SERVICES

S3 — Simple Storage Service

S3 is AWS's object storage service — store any amount of data, retrieve it from anywhere. Objects are stored in Buckets. S3 provides 99.999999999% (11 nines) of durability.

S3 Key Concepts	
Bucket	→ Container for objects (globally unique name, per-region)
Object	→ File + metadata, up to 5 TB per object
Key	→ Full path of the object (folder/subfolder/filename.txt)
URL	→ https://bucket-name.s3.amazonaws.com/key

S3 Storage Classes

Class	Use Case	Retrieval
S3 Standard	Frequently accessed data	Milliseconds
S3 Intelligent-Tiering	Unknown access patterns	Milliseconds
S3 Standard-IA	Infrequently accessed	Milliseconds
S3 One Zone-IA	Non-critical, infrequent access	Milliseconds
S3 Glacier Instant	Archive, accessed occasionally	Milliseconds
S3 Glacier Flexible	Archive, retrieval in minutes-hours	Minutes-Hours
S3 Glacier Deep Archive	Long-term archive, accessed rarely	Hours

EBS — Elastic Block Store

EBS provides persistent block storage volumes for EC2 instances. Like an external hard drive that attaches to your virtual server. EBS volumes persist independently of the EC2 instance lifecycle.

EBS Volume Type	Best For
gp3 (General Purpose SSD)	Most workloads — boot volumes, dev/test
io2 Block Express	Highest performance — large databases
st1 (Throughput HDD)	Big data, log processing, data warehouses
sc1 (Cold HDD)	Lowest cost, infrequently accessed data

EFS — Elastic File System

EFS is a managed NFS (Network File System) that can be mounted concurrently by thousands of EC2 instances across multiple AZs. Automatically scales — no capacity planning needed.

Feature	EBS	EFS
Type	Block Storage	File System (NFS)
Access	1 EC2 at a time	1000s of EC2s simultaneously
Scalability	Fixed size (provision)	Automatic, elastic
AZ	Single AZ	Multi-AZ
Use Case	Boot volumes, DBs	Shared file storage, CMS

FSx & Storage Gateway

Service	Description
FSx for Windows	Managed Windows File Server — SMB protocol, Active Directory
FSx for Lustre	High-performance file system for HPC, ML, media processing
Storage Gateway	Hybrid: connects on-premises to AWS S3/EBS/FSx via iSCSI/NFS/SMB

tech__talks

7 Chapter 7

DATABASE SERVICES

RDS — Relational Database Service

RDS is a managed relational database service. AWS handles provisioning, patching, backup, recovery, and scaling — you focus on data and queries.

RDS Supported Engines

MySQL • PostgreSQL • MariaDB • Oracle • SQL Server • IBM Db2
Multi-AZ: Standby replica in another AZ for automatic failover (HA)
Read Replicas: Read-only copies to scale read traffic (up to 5 replicas)
Automated Backups: Daily snapshot + transaction logs, retention 0-35 days

DynamoDB

DynamoDB is a fully managed, serverless NoSQL key-value and document database. Single-digit millisecond performance at any scale. No servers to manage — fully serverless.

- Tables: Collection of items (rows). No fixed schema
- Primary Key: Partition Key alone, or Partition Key + Sort Key
- Capacity: On-demand (pay per request) or Provisioned (set read/write units)
- DynamoDB Streams: Capture change events for triggering Lambda
- Global Tables: Multi-region, active-active replication

Aurora

Aurora is AWS's proprietary cloud-native relational database — compatible with MySQL and PostgreSQL, but 3-5x faster. Storage auto-scales up to 128 TB. 6 copies of data across 3 AZs.

Aurora Feature	Detail
Performance	5x MySQL, 3x PostgreSQL speed
Storage	Auto-scales 10 GB to 128 TB
Replicas	Up to 15 Aurora Replicas
Aurora Serverless	Auto-scales compute — ideal for intermittent workloads
Global Database	Replication across 5 regions, <1 second lag

Other Database Services

Service	Purpose
Redshift	Data warehouse — petabyte-scale analytics using SQL

ElastiCache	In-memory caching — Redis or Memcached — sub-millisecond latency
DocumentDB	MongoDB-compatible managed document database
Neptune	Managed graph database for relationship-heavy data
Keyspaces	Managed Apache Cassandra (wide-column)
MemoryDB for Redis	Redis-compatible, durable, ultra-fast, primary database

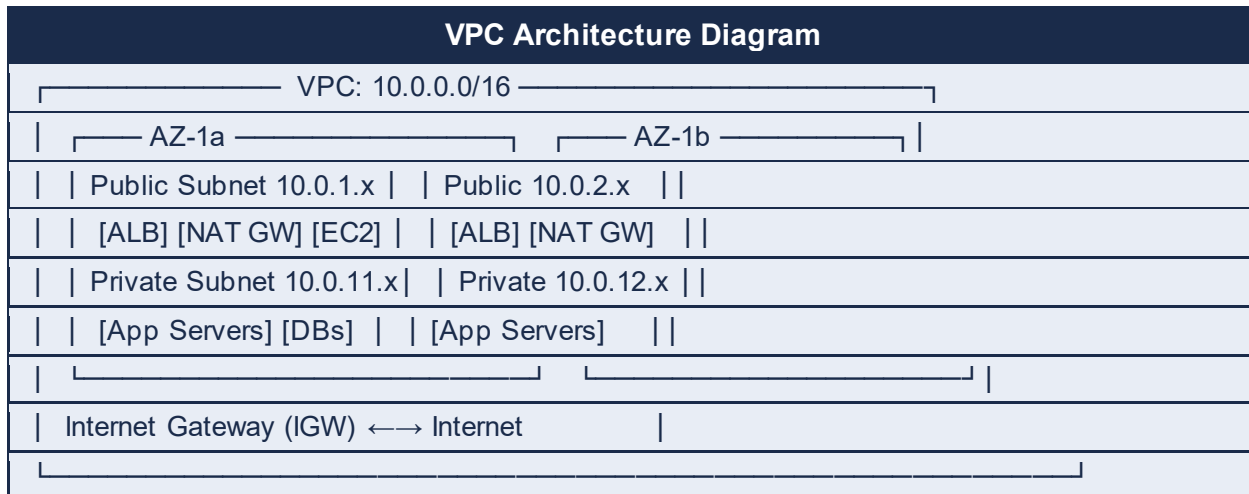
Database Backup & Replication

- Automated Backups: RDS takes daily snapshots + continuous transaction logs
- Manual Snapshots: User-initiated, retained until deleted
- Cross-Region Snapshots: Copy snapshots to another region for DR
- Read Replicas: Offload read traffic; can be promoted to standalone DB
- Multi-AZ: Synchronous replication for HA (not for read scaling)

tech__talks

8 Chapter 8**NETWORKING & CONTENT DELIVERY****VPC — Virtual Private Cloud**

A VPC is your private, logically isolated network within AWS. You define IP ranges, subnets, route tables, and gateways. Think of it as your own data center inside AWS.

**Subnets**

Subnets partition your VPC's IP space. They exist in a single AZ.

Subnet Type	Description
Public Subnet	Has route to Internet Gateway — resources get public IPs
Private Subnet	No direct internet route — uses NAT Gateway for outbound only

Route Tables

Route tables control where network traffic from subnets is directed. Each subnet is associated with one route table.

- Public RT: 0.0.0.0/0 → Internet Gateway (allows internet traffic)
- Private RT: 0.0.0.0/0 → NAT Gateway (outbound only, no inbound)

Internet Gateway & NAT Gateway

Component	Purpose
Internet Gateway (IGW)	Allows public subnets to communicate with the Internet (bidirectional)
NAT Gateway	Allows private subnets to initiate outbound internet traffic (no inbound)
Egress-Only IGW	IPv6 equivalent of NAT — outbound IPv6 only

Security Groups & NACLs

Feature	Security Group	NACL
Level	Instance level	Subnet level
State	Stateful	Stateless (return traffic must be explicitly allowed)
Rules	Allow only	Allow and Deny
Applies	EC2, RDS, Lambda, etc	All traffic in/out of subnet

Route 53

Route 53 is AWS's scalable DNS (Domain Name System) and domain registrar. It routes users to applications using various routing policies.

Routing Policy	Use Case
Simple	Single resource — no health checks
Weighted	A/B testing — split traffic 80/20
Latency	Route to lowest-latency region
Failover	Active-passive — failover on health check failure
Geolocation	Route based on user's country/continent
Multi-value	Returns multiple healthy IPs — basic load balancing

CloudFront — CDN

CloudFront is AWS's Content Delivery Network. It caches content at 400+ edge locations globally, reducing latency for users worldwide. Integrates with S3, ALB, EC2, and API Gateway.

- Origins: S3 bucket, ALB, EC2, API Gateway, HTTP server
- Behaviors: Cache rules per URL path — /api/* vs /static/*
- OAC: Origin Access Control — CloudFront-only S3 access (no public S3)
- HTTPS: Free SSL/TLS via ACM certificates

tech__talks

9 Chapter 9

VPC ADVANCED CONCEPTS

VPC Peering

VPC Peering connects two VPCs privately using AWS's network — traffic never traverses the public internet. Works within and across accounts and regions.

- Not transitive: $A \leftrightarrow B$ and $B \leftrightarrow C$ does NOT mean $A \leftrightarrow C$
- No overlapping CIDR blocks allowed
- Must update route tables in both VPCs

Transit Gateway

Transit Gateway acts as a central hub connecting thousands of VPCs, VPNs, and Direct Connect connections. Solves the complexity of many VPC peering connections.

Transit Gateway Hub Model	
VPC-A	—
VPC-B	—
VPC-C	— —▶ Transit Gateway ◀— VPN (on-premises)
VPC-D	— ◀— Direct Connect
VPC-E	—
All VPCs can communicate via TGW — one route table to manage	

VPN & Direct Connect

Connection	Description
Site-to-Site VPN	Encrypted IPSec tunnel over public internet — fast setup, variable performance
Direct Connect	Dedicated private physical link to AWS — consistent bandwidth, higher cost
VPN + Direct Connect	Use VPN as backup to Direct Connect for HA

PrivateLink & Endpoint Services

VPC Endpoints allow private connectivity to AWS services without internet, NAT, or VPN. Traffic stays within the AWS network.

Endpoint Type	Description
Gateway Endpoint	For S3 and DynamoDB — free, route table entry
Interface Endpoint	For most other services — ENI with private IP (costs money)
PrivateLink	Expose your own service privately to other VPCs/accounts

 *Exam: If the question says 'communicate with S3 without going through the internet', the answer is VPC Gateway Endpoint (free).*

[tech__talks](#)

10 Chapter 10

MONITORING & LOGGING

CloudWatch

CloudWatch is AWS's observability service for metrics, logs, alarms, and dashboards. Nearly every AWS service publishes metrics to CloudWatch automatically.

CloudWatch Feature	Description
Metrics	Numeric data over time — CPU, NetworkIn, RequestCount
Logs	Log groups from EC2, Lambda, VPC Flow Logs, API GW
Alarms	Trigger actions when metric crosses threshold
Dashboards	Custom real-time visualizations of metrics
Events/Rules	Schedule actions or respond to AWS events
Insights	Query log data with CloudWatch Logs Insights (SQL-like)

CloudTrail

CloudTrail records every API call made in your AWS account — who did what, when, and from where. Essential for auditing, compliance, and security investigation.

- Records: Console actions, CLI commands, SDK calls, service-to-service calls
- Event types: Management events (control plane), Data events (S3/Lambda)
- Trail: Configure to deliver events to S3 + CloudWatch Logs
- Insight Events: Detect unusual API activity automatically

AWS Config

AWS Config continuously records configuration changes to AWS resources and evaluates them against desired compliance rules.

- Tracks: Who changed what resource, when, and what it changed from/to
- Rules: Pre-built or custom Lambda-backed rules (e.g., 'S3 must not be public')
- Remediation: Auto-remediate non-compliant resources via SSM

X-Ray

AWS X-Ray provides distributed tracing for applications. Visualizes request flows through microservices, identifies bottlenecks, and pinpoints error sources.

- Service Map: Visual graph of all services a request touches
- Traces: End-to-end request journey with timing per service
- Segments & Subsegments: Fine-grained timing within a service

tech__talks

1 1 Chapter 11**SECURITY & COMPLIANCE****Shared Responsibility Model**

AWS and customers share security responsibility. AWS secures the infrastructure; you secure what you put on it.

AWS Responsibility (Security OF the cloud)	Customer Responsibility (Security IN the cloud)
Physical data centers, hardware	Data encryption at rest and in transit
Networking infrastructure	OS patching on EC2 instances
Hypervisor and virtualization	IAM users, roles, policies
Managed service internals (RDS OS)	Security groups and NACLs
Global infrastructure	Application-level security

Key Security Services

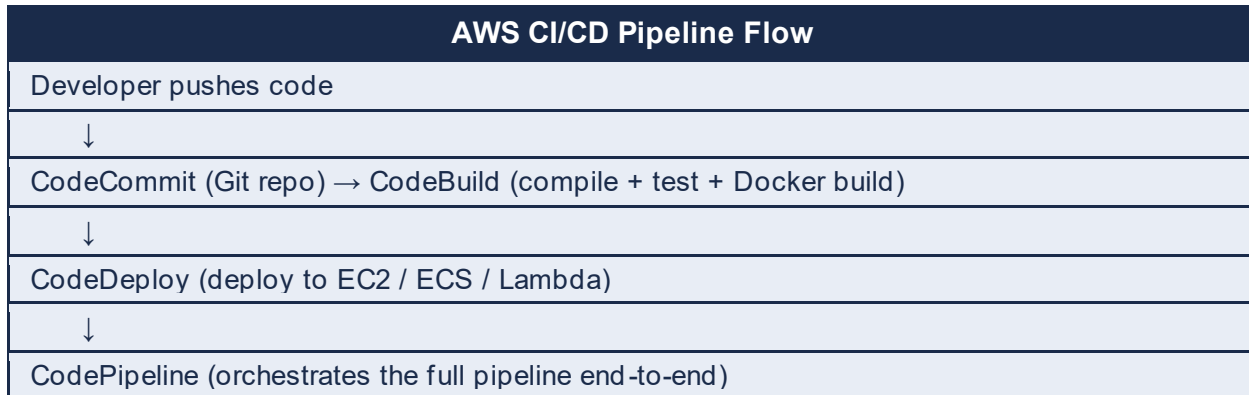
Service	Purpose
KMS (Key Management Service)	Create and manage encryption keys for data at rest
Secrets Manager	Store, rotate, and retrieve credentials/API keys securely
Parameter Store (SSM)	Store config, secrets, and parameters — free tier available
WAF (Web Application Firewall)	Block SQL injection, XSS, rate limiting at HTTP layer
Shield Standard	Free DDoS protection for all AWS customers
Shield Advanced	Enhanced DDoS — 24/7 DRT team, cost protection
GuardDuty	ML-powered threat detection — analyzes CloudTrail, VPC logs, DNS
Inspector	Automated vulnerability scanning for EC2 and container images
Macie	ML-powered PII discovery and data security in S3

tech__talks

1 2 Chapter 12

DEVOPS & DEVELOPER TOOLS

AWS Developer Tools — CI/CD Pipeline



Service	Role in CI/CD
CodeCommit	Managed private Git repository (like GitHub, hosted in AWS)
CodeBuild	Fully managed build service — compiles, runs tests, produces artifacts
CodeDeploy	Automated deployment to EC2, Lambda, ECS, and on-premises
CodePipeline	Orchestrator — chains Source → Build → Test → Deploy stages

CloudFormation

CloudFormation is AWS's native Infrastructure as Code service. You define resources in YAML or JSON templates, and CloudFormation provisions and manages them as stacks.

- Template: YAML/JSON file describing resources (EC2, VPC, RDS, etc.)
- Stack: Running instance of a template — resources created/updated/deleted together
- Change Set: Preview changes before applying — like 'terraform plan'
- Drift Detection: Detect manual changes to stack resources

CDK — Cloud Development Kit

CDK lets you define AWS infrastructure using familiar programming languages: TypeScript, Python, Java, C#, Go. CDK synthesizes your code into CloudFormation templates.

Elastic Beanstalk

Elastic Beanstalk is a PaaS service — you upload your code and Beanstalk handles the deployment, capacity provisioning, load balancing, auto-scaling, and health monitoring.

- Supports: Node.js, Python, Java, Ruby, PHP, Go, .NET, Docker

- You control: Code, environment config; Beanstalk manages the rest

 *Beanstalk is free — you pay only for underlying EC2, RDS, ALB resources.*

tech__talks

1 3 Chapter 13

CONTAINERS & KUBERNETES

Docker on AWS

Containers package an application with all its dependencies into a portable unit. AWS provides multiple services for running, storing, and orchestrating containers.

Container Services on AWS	
ECR (Elastic Container Registry) → Store Docker images	
↓	↓
ECS (AWS-native orchestration)	EKS (Kubernetes orchestration)
↓	↓
EC2 launch type (you manage EC2) OR Fargate (serverless, no EC2)	

Service	Description
ECR	Managed Docker/OCI container registry. Secure, versioned image storage
ECS	AWS-native orchestrator. Simpler than Kubernetes. Task = unit of work
EKS	Managed Kubernetes control plane. Run standard K8s workloads
Fargate	Serverless container compute. No EC2 to manage. Pay per task/pod CPU+memory

ECS vs EKS — When to Use Which

Choose ECS if...	Choose EKS if...
You're new to containers	You already use Kubernetes
You want simpler operations	You need Kubernetes-specific features
AWS-native integration is preferred	Multi-cloud or hybrid Kubernetes required
Smaller team, less K8s expertise	Large team with K8s knowledge

tech__talks

Chapter 14

INFRASTRUCTURE AS CODE (IAC)

IaC Overview

Infrastructure as Code means defining and provisioning infrastructure using code files instead of manual clicks. IaC enables version control, repeatability, peer review, and automation.

IaC Tool	Language / Approach
CloudFormation	AWS-native, YAML/JSON templates, free
Terraform	HCL (HashiCorp Config Language), multi-cloud, state-based
CDK	TypeScript/Python/Java/Go, generates CloudFormation
Pulumi	General-purpose languages (Python, TypeScript, Go)

CloudFormation Deep Dive

Template Structure

CloudFormation Template Sections
AWSTemplateFormatVersion: '2010-09-09'
Description: My stack description
Parameters: (inputs — like variables)
Mappings: (key-value lookup tables)
Conditions: (if/else logic)
Resources: (REQUIRED — the actual AWS resources)
Outputs: (values to export for other stacks)

Terraform with AWS

Terraform uses HCL to declare the desired state. It compares desired vs actual state, builds a plan, and applies only the differences.

Terraform Concept	Meaning
Provider	Plugin for a cloud — aws, azure, google
Resource	aws_instance, aws_vpc, aws_s3_bucket
State file	terraform.tfstate — records current reality
Plan	terraform plan — shows what will change
Apply	terraform apply — makes the changes
Destroy	terraform destroy — removes all resources
Modules	Reusable packages of Terraform resources
Remote State	Store state in S3 + lock with DynamoDB for teams

CDK — Cloud Development Kit

CDK uses constructs (L1, L2, L3) to abstract CloudFormation complexity. L1 = raw CFN resource, L2 = opinionated resource with sensible defaults, L3 = complete patterns.

tech__talks

1 5 Chapter 15

SERVERLESS ARCHITECTURE

Serverless Overview

Serverless means you write code and AWS manages all servers, scaling, and availability. You pay only for execution — no idle costs.

Serverless Request Flow

Client → API Gateway → Lambda → DynamoDB

↓

EventBridge → Step Functions → Lambda chain

Analogy: Like a taxi (serverless) vs owning a car (EC2) — pay per ride, no maintenance.

Service	Role in Serverless
Lambda	Compute — run code in response to events
API Gateway	Create REST/HTTP/WebSocket APIs that trigger Lambda
DynamoDB	Serverless NoSQL database with auto-scaling
Step Functions	Orchestrate multi-step workflows as state machines
EventBridge	Event bus — route events between services and third parties
S3	Store static files, trigger Lambda on object upload
SQS	Decouple services, trigger Lambda from queue messages

API Gateway

API Gateway creates, publishes, maintains, monitors, and secures APIs at scale. Supports REST API, HTTP API (cheaper, faster), and WebSocket API.

- Authentication: IAM, Cognito, Lambda Authorizer (custom auth)
- Rate limiting: Throttle requests per API key or stage
- Stages: dev, staging, prod — deploy to different environments
- Caching: Cache responses to reduce backend calls

Step Functions

Step Functions orchestrates complex workflows as visual state machines. Each step can be a Lambda, ECS task, DynamoDB call, SNS publish, or manual approval.

- Standard Workflow: Long-running (up to 1 year), exactly-once execution
- Express Workflow: High-volume, short-duration (up to 5 min), at-least-once

tech__talks

1 6 Chapter 16

MESSAGING & INTEGRATION

Messaging Services Overview

Messaging services decouple application components, enabling asynchronous communication and event-driven architectures.

Service	Type & Use Case
SQS	Queue — async, decoupled, pull-based. Buffer between producers and consumers
SNS	Pub/Sub — push notifications to many subscribers simultaneously
EventBridge	Event bus — route events from AWS/SaaS/custom apps to targets
MQ	Managed ActiveMQ/RabbitMQ — migrate existing message broker workloads
Kinesis	Real-time data streaming — logs, clickstreams, IoT telemetry

SQS — Simple Queue Service

Feature	Detail
Standard Queue	At-least-once delivery, best-effort ordering, unlimited throughput
FIFO Queue	Exactly-once, strict order, 3000 msg/sec
Visibility Timeout	Message hidden during processing — prevent double processing
Dead Letter Queue	Failed messages after max retries go to DLQ for debugging
Long Polling	Wait up to 20s for messages — reduces empty responses and cost

SNS — Simple Notification Service

SNS uses a pub/sub model. Publishers send to a Topic; Subscribers (SQS, Lambda, email, HTTP, SMS, mobile push) receive all messages from that topic.

SNS Fan-Out Pattern
Publisher → SNS Topic → SQS Queue A (process orders)
→ SQS Queue B (send email)
→ Lambda (update inventory)
Fan-Out: One event triggers multiple downstream processes simultaneously

Kinesis

Kinesis Service	Purpose
-----------------	---------

Kinesis Data Streams	Ingest real-time data, store for up to 365 days, process with Lambda/KCL
Kinesis Data Firehose	Load streaming data to S3, Redshift, OpenSearch — fully managed
Kinesis Data Analytics	SQL or Flink queries on streaming data in real time
Kinesis Video Streams	Ingest, store, and process video streams

tech__talks

17 Chapter 17

BACKUP & DISASTER RECOVERY

AWS Backup

AWS Backup is a centralized, policy-driven backup service for EC2, EBS, RDS, DynamoDB, EFS, FSx, and Storage Gateway. Define backup plans with retention policies and vaults.

DR Strategies — RTO vs RPO

Term	Definition
RTO (Recovery Time Objective)	Max acceptable downtime — how quickly must you recover?
RPO (Recovery Point Objective)	Max acceptable data loss — how recent must the backup be?

DR Strategies (Cost vs Recovery Speed)		
Backup & Restore	← Cheapest, slowest (hours)	RTO: Hours, RPO: Hours
Pilot Light	← Core services running,	RTO: 10 min, RPO: Minutes
scale up on failure		
Warm Standby	← Scaled-down live copy,	RTO: Minutes, RPO: Seconds
promote on failure		
Multi-Site Active	← Full copy in 2+ regions,	RTO: Real-time, RPO: Zero
traffic splits immediately		

Strategy	Description
Backup & Restore	S3 backups + AMI snapshots. Restore and rebuild on failure. Cheapest.
Pilot Light	Core DB replica + minimal app in DR region. Scale compute on event.
Warm Standby	Full app in DR at reduced scale. Already running — just scale up.
Multi-Site	Full production in 2+ regions. Route 53 failover routes instantly.

Cross-Region Replication

- S3 CRR: Replicate S3 objects to another region automatically
- RDS Cross-Region Read Replicas: Can be promoted during DR
- DynamoDB Global Tables: Multi-region active-active replication
- EBS Snapshots: Copy snapshots cross-region for EC2 DR

tech__talks

18 Chapter 18

COST MANAGEMENT

EC2 Pricing Models

Pricing Model	Description
On-Demand	Pay per second/hour. No commitment. Most flexible, highest price.
Reserved Instances	1 or 3-year commitment. Up to 72% discount. For steady workloads.
Savings Plans	Flexible RI — commit to \$/hour spend. Covers EC2, Lambda, Fargate.
Spot Instances	Bid for spare capacity. Up to 90% discount. Can be interrupted.
Dedicated Hosts	Physical server dedicated to you. For compliance/licensing needs.
Dedicated Instances	Your instances, isolated hardware. Less control than Dedicated Host.

Cost Optimization Tools

Tool	Purpose
Cost Explorer	Visualize, analyze, and forecast spend. Rightsizing recommendations.
AWS Budgets	Set cost/usage/RI coverage alerts. Email or SNS when exceeded.
Cost & Usage Report	Granular line-item report → S3 → Athena for deep analysis.
Compute Optimizer	ML recommendations for EC2, Lambda, EBS right-sizing.
Trusted Advisor	Best practice checks: cost, security, fault tolerance, performance.
Cost Allocation Tags	Tag resources (Team, Project) → filter costs in Cost Explorer.

 **Savings Plans vs Reserved Instances:** Savings Plans offer more flexibility (cover EC2 + Lambda + Fargate across regions/families). Reserved Instances are locked to a specific instance family and region.

tech__talks

1 9 Chapter 19**HIGH AVAILABILITY & SCALABILITY****High Availability (HA)**

High Availability means designing systems to be continuously operational, minimizing downtime. AWS achieves HA by deploying across multiple AZs and Regions.

HA Architecture Pattern	
Route 53 (DNS + Health Check)	
↓	
ALB (spans AZ-a + AZ-b)	
↓ ↓	
ASG: EC2 (AZ-a) ASG: EC2 (AZ-b)	
↓ ↓	
RDS Primary (AZ-a) ←sync→ RDS Standby (AZ-b) [Auto failover]	

Scalability Types

Type	Description
Vertical Scaling (Scale Up)	Increase instance size (t3.micro → m5.large). Has limits.
Horizontal Scaling (Scale Out)	Add more instances. Unlimited scale. Preferred in cloud.

Auto Scaling Group Policies

Policy Type	How It Works
Target Tracking	Maintain a target metric — e.g., keep CPU at 60%
Step Scaling	Add/remove instances in steps based on alarm thresholds
Scheduled	Pre-defined scale times — e.g., scale up at 8 AM weekdays
Predictive	ML forecasts traffic, pre-scales before demand spike

Fault Tolerance vs High Availability

Concept	Meaning
High Availability	Minimize downtime — fast recovery. Some downtime acceptable.
Fault Tolerance	Zero downtime — system continues through component failures.
Disaster Recovery	Restore service after major failure or outage.

tech__talks

20 Chapter 20

MIGRATION & TRANSFER

Migration Services

Service	Use Case
AWS Migration Hub	Central dashboard to track migration progress across tools
DMS (Database Migration Service)	Migrate databases to AWS — homogeneous and heterogeneous
Server Migration Service (SMS)	Migrate on-premises VMs to AWS incrementally
Application Discovery Service	Discover on-premises servers, dependencies, and performance

Data Transfer — Snow Family

When network transfer is too slow or expensive, AWS ships physical devices to move terabytes or petabytes of data offline.

Device	Capacity	Use Case
Snowcone	8-14 TB	Edge compute, small transfers, fits in a backpack
Snowball Edge	80 TB	Large migrations, edge compute, remote office
Snowmobile	100 PB	Exabyte-scale migration — a literal shipping truck

DataSync

DataSync is an online data transfer service to move large amounts of data between on-premises storage and AWS (S3, EFS, FSx). Up to 10x faster than open-source tools.

The 6 Rs of Migration

Strategy	Description
Rehost (Lift & Shift)	Move as-is to EC2. Fastest. No optimization.
Replatform (Lift & Tinker)	Minor optimizations — move DB to RDS, app to Beanstalk
Repurchase	Move to SaaS — e.g., replace CRM with Salesforce
Refactor/Re-architect	Redesign for cloud-native — microservices, serverless
Retire	Decommission workloads no longer needed
Retain	Keep on-premises for now (compliance, dependencies)

tech__talks

2 1 Chapter 21

ANALYTICS & BIG DATA

Service	Description
Athena	Serverless SQL query engine on S3 data. Pay per query. No ETL needed.
EMR	Managed Hadoop/Spark/Hive cluster. Process petabytes of data.
Glue	Serverless ETL (Extract, Transform, Load) + Data Catalog.
QuickSight	Business intelligence (BI) — dashboards, visualizations, ML insights.
Lake Formation	Build secure data lakes on S3 with fine-grained access control.
OpenSearch	Search and analytics (Elasticsearch successor) — logs, full-text search.

Modern Data Lake Architecture on AWS	
Data Sources (RDS, IoT, S3 uploads, Kinesis streams)	
↓	
AWS Glue (ETL) → S3 Data Lake (raw / processed / curated zones)	
↓	↓
Glue Data Catalog	Athena (SQL queries)
↓	
QuickSight (dashboards) / Redshift (data warehouse)	

tech__talks

2 2 Chapter 22

AI & ML SERVICES (BASICS)

AWS AI/ML Service Stack

Service	What It Does
SageMaker	Build, train, and deploy ML models. Full ML lifecycle management.
Rekognition	Image and video analysis — detect objects, faces, text, scenes
Comprehend	NLP — sentiment analysis, entity detection, language detection
Lex	Build conversational chatbots (same tech as Alexa)
Polly	Text-to-speech — convert text to natural-sounding audio
Transcribe	Speech-to-text — transcribe audio/video files automatically
Translate	Neural machine translation — 75+ languages
Textract	Extract text and data from scanned documents and PDFs
Forecast	Time-series forecasting using ML
Personalize	Real-time personalization and recommendations (like Amazon.com)
Bedrock	Access foundation models (Claude, Llama, Titan) via API

💡 *SageMaker Studio is a web IDE for the full ML workflow: data prep → training → evaluation → deployment → monitoring.*

tech__talks

2 3 Chapter 23**AWS WELL-ARCHITECTED FRAMEWORK****The 6 Pillars**

The Well-Architected Framework provides best practices for designing and operating reliable, secure, efficient, cost-effective, and sustainable systems in the cloud.

Pillar	Focus	Key Principle
Operational Excellence	Run & monitor systems	Automate operations, make frequent small reversible changes
Security	Protect data & systems	Apply security at all layers, least privilege, encrypt everything
Reliability	Recover from failures	Automatically recover, scale horizontally, test recovery
Performance Efficiency	Use resources efficiently	Use serverless, experiment often, go global in minutes
Cost Optimization	Avoid unnecessary costs	Pay for what you use, right-size, use Spot where possible
Sustainability	Minimize environmental impact	Maximize utilization, use managed services, reduce waste

 Use the AWS Well-Architected Tool (free) to review workloads against the 6 pillars and get recommendations.

tech__talks

2 4 Chapter 24**AWS REAL-TIME USE CASES****1. Web Application Hosting**

3-Tier Web App on AWS
Users → Route 53 → CloudFront (CDN)
↓
ALB (multi-AZ) → EC2 Auto Scaling Group (private subnets)
↓
RDS Multi-AZ (PostgreSQL) + ElastiCache (Redis session cache)
S3 (static assets) + CloudWatch (monitoring) + CloudTrail (audit)

2. CI/CD Pipeline

Automated Deployment Pipeline
GitHub/CodeCommit push → CodePipeline triggers
→ CodeBuild (unit tests + Docker build + ECR push)
→ CodeDeploy (Blue/Green deploy to ECS Fargate)
→ CloudWatch alarms monitor new deployment
→ Rollback automatic if error rate > 5%

3. Microservices Architecture

Microservices on ECS + API Gateway
Client → API Gateway → Lambda Authorizer (JWT validation)
↓
ALB → ECS Fargate services:
- User Service → RDS PostgreSQL
- Order Service → DynamoDB
- Notification Service → SNS → SQS → Lambda
X-Ray traces all service-to-service calls

4. Serverless Application

Fully Serverless Stack
React SPA → S3 + CloudFront (hosting)
API calls → API Gateway → Lambda → DynamoDB

File uploads → S3 → Lambda (resize image) → S3 (thumbnail)
Async jobs → SQS → Lambda (process) → SNS (notify user)
Auth → Cognito User Pools + Identity Pools

5. Big Data Processing

Real-Time Analytics Pipeline

IoT sensors / App events → Kinesis Data Streams
→ Lambda (real-time alerts) + Kinesis Firehose (batch)
→ S3 (data lake, raw zone)
→ Glue ETL → S3 (processed zone) + Glue Data Catalog
→ Athena (ad-hoc SQL) + Redshift (warehouse) + QuickSight (viz)

6. DevOps Automation

Infrastructure Automation

Terraform / CDK → CloudFormation → AWS Resources
GitHub Actions → CodePipeline → CodeBuild → ECR
Config changes → AWS Config Rules → Lambda auto-remediation
Alerts → CloudWatch Alarm → SNS → PagerDuty / Slack
Secrets → Secrets Manager (auto-rotate) → Lambda / ECS

tech__talks

AWS Cloud Complete Textbook

24 Chapters | 200+ Services | Diagrams | Best Practices

tech__talks